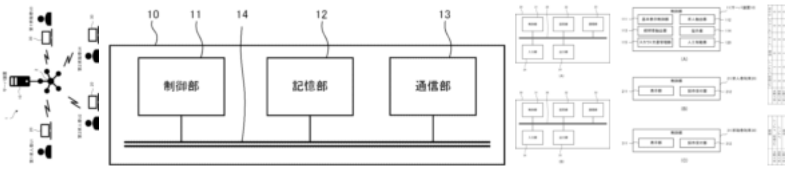
Recruitment support system, recruitment support method and program

Abstract

translated from Japanese

[Problem] To provide a recruitment support system, method, and program for presenting combinations of job offers and candidates to employers. [Solution] In the recruitment support system (1), a job offer extraction step executed by a control unit of a server device extracts at least one recommended job offer from among the job offers registered in a database by employers, based on the content of the job offer and an expected value for success calculated based on first reference information. The first reference information includes a correlation between the content of the job offer and the expected value for success, which is the probability of success for the job offer. In the candidate extraction step, one or more candidates for the recommended job offer are extracted from among job seekers registered in the database, based on the content of the recommended job offer and a job seeker matching index calculated based on second reference information. The second reference information includes a correlation between the content of the job offer and a job seeker matching index indicating the degree of suitability of the job seeker for the job offer. In the presentation step, the recommended job offers and candidates are associated and presented to the employer. [Selected Figure] FIG. 1

Images (8)



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Inventor: 要一 原田

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Claims (17)

Hide Dependent

translated from Japanese

A recruitment support system,
A processor is provided.
The processor is configured to perform the steps of:
In the job offer extraction step, at least one recommended job offer is extracted from the job offers registered in the database by the job recruiter based on the content of the job offer and a contract expectation value calculated based on the first reference information, where the first reference information includes a correlation between the content of the job offer and the contract expectation value, which is a probability of the job offer being concluded, and
In the candidate extraction step, at least one candidate for the recommended job offer is extracted from among job seekers registered in the database, based on a job seeker matching index calculated based on the content of the recommended job offer and second reference information, wherein the second reference information includes a correlation between the content of the job offer and the job seeker matching index indicating a degree of suitability of the job seeker for the job offer;
In the presenting step, the recommended job offers are associated with the candidates and presented to the recruiter ;
In the scout document management step, the recruiter is given a stock of scout emails that can be sent only to the candidate, or the recruiter's stock of scout emails is not consumed when the recruiter sends a scout email to the candidate, in this recruitment support system. A recruitment support system,
A processor is provided.
The processor is configured to perform the steps of:
In the job offer extraction step, at least one recommended job offer is extracted from the job offers registered in the database by the job recruiter based on the content of the job offer and a contract expectation value calculated based on the first reference information, where the first reference information includes a correlation between the content of the job offer and the contract expectation value, which is a probability of the job offer being concluded, and
In the candidate extraction step, at least one candidate for the recommended job offer is extracted from among job seekers registered in a database, based on a job seeker matching index calculated based on the content of the recommended job offer and second reference information, wherein the second reference information includes information for calculating, as the job seeker matching index, a degree of similarity between a job seeker for whom the job seeker matching index is to be calculated and a job seeker who has a history of actions related to a job offer similar to the recommended job offer;
In the presentation step, the recruitment support system presents the recommended job offers and the candidates in association with each other to the recruiter. 3. The recruitment support system according to claim 1,
The first reference information is an expected value calculation model that is trained to be able to input the content of a job offer and output the expected value of the contract,
In the job offer extraction step, the contents of the job offer are input to the expectation value calculation model, and the expectation value calculation model is caused to output the contract expectation value, in the recruitment support system. 4. The recruitment support system according to claim 3 ,
The employment support system, wherein the expected value calculation model is learned using the contents of a plurality of job offers and whether or not these job offers have been concluded within a predetermined period of time. 3. The recruitment support system according to claim 1,
In the job extraction step, a job offer whose expected value of contract is equal to or greater than a predetermined threshold is extracted as the recommended job offer. 3. The recruitment support system according to claim 1,
In the job offer extraction step, the recruitment support system extracts recommended job offers from the job offers registered in the database by the employer based on the

expected value of the job offer and the history of sending scout documents based on the job offer to job seekers. 7. The recruitment support system according to claim 6 ,
In the job offer extraction step, the recommended job offers are extracted based on a job offer score calculated for each job offer,
The job offer score increases according to the magnitude of the expected value of the contract and its weighting, and decreases according to the magnitude of the number of transmissions of the scout document in a predetermined period of time and its weighting,
A recruitment support system, wherein the weighting of the expected value of contract is greater than the weighting of the number of transmissions. The recruitment support system according to claim 7 ,
In the job offer extraction step, a first job offer score and a second job offer score, in which at least one of the expected value for contract and the number of transmissions is weighted differently, are calculated for each job offer as the job offer score, and at least one first recommended job offer extracted based on the first job offer score and at least one second recommended job offer extracted based on the second job offer score are extracted as the recommended job offers;
In the presenting step, the first recommended job offer and the second recommended job offer are presented to the recruiter. 3. The recruitment support system according to claim 1,
the second reference information is a matching index calculation model that is trained to be able to input the content of a job offer and output the job seeker matching index for each job seeker,
In the candidate extraction step, the contents of the recommended job offers are input to the matching index calculation model, and the matching index calculation model is caused to output the job seeker matching index. The recruitment support system according to claim 9 ,
The matching index calculation model is a learning model that learns the relationship between the contents of multiple job openings and job seekers who have a history of actions related to the job openings, and outputs the job seeker matching index as the degree of similarity between the input job opening and a job seeker who has a history of actions related to a similar job opening. The recruitment support system according to claim 10 ,
The action is receiving a scout document or passing an examination, in the recruitment support system. 3. The recruitment support system according to claim 1,
In the candidate extraction step, the recruitment support system extracts the candidates from among job seekers registered in a database based on the job seeker matching index of the job seeker and the number of activities of the job seeker for any job offer. 3. The recruitment support system according to claim 1,
In the presenting step, a portion of the recommended job offers is selected from the extracted plurality of recommended job offers for each predetermined period and presented. 3. The recruitment support system according to claim 1,
In the presenting step, a portion of the candidates is selected from the extracted plurality of candidates for each predetermined period and presented. 3. The recruitment support system according to claim 1,
In the presenting step, a label corresponding to the expected value of contract is assigned to the recommended job offer and presented. A recruitment support method, comprising:
A recruitment support method comprising the steps executed by the recruitment support system according to claim 1 or 2 . A program,
A program for causing a computer to execute each step of the recruitment support system according to claim 1 or 2 .

Description

translated from Japanese

The present invention relates to a recruitment support system, a recruitment support method, and a program.

As disclosed in Patent Document 1, a technology is known that allows employers to search for job seekers who meet the desired criteria based on job seeker information registered by the job seeker.

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With this type of technology, employers need to input keywords and other criteria to search for the job seekers they are looking for.

In consideration of the above circumstances, the present invention provides a recruitment support system that can present combinations of job openings and candidates to employers.

According to one aspect of the present invention, a recruitment support system is provided. The recruitment support system includes a processor. The processor is configured to execute the following steps: In the job offer extraction step, at least one recommended job offer is extracted from the job offers registered in the database by the employer, based on the content of the job offer and a closing expectation value calculated based on the first reference information. The first reference information includes a correlation between the content of the job offer and the closing expectation value, which is the probability of closing the job offer for the job offer. In the candidate extraction step, at least one candidate for the recommended job offer is extracted from the job seekers registered in the database, based on the content of the recommended job offer and a job seeker matching index calculated based on the second reference information. The second reference information includes a correlation between the content of the job offer and a job seeker matching index indicating the degree of suitability of the job seeker for the job offer. In the presentation step, the recommended job offer and the candidate are presented to the employer in association with each other.

According to this embodiment, recommended job offers with a high expected success rate can be presented to the employer along with candidates who are highly suitable for the recommended job offers. Therefore, the employer, who is the user, can increase the possibility of the job offer being successful by taking an action based on the recommended job offer (e.g., sending a scout document). In addition, by presenting candidates for the recommended job offers, the employer can identify job seekers who are likely to succeed in the job offer.

1 is a configuration diagram showing a recruitment support system 1. FIG. 2 is a block diagram showing the hardware configuration of the server device 10. FIG. 2 is a block diagram showing the hardware configuration of a recruiter terminal 20 and a job seeker terminal 30. 2 is a block diagram showing functions realized by the server device 10 (control unit 11), the recruiter terminal 20 (control unit 21), and the job seeker terminal 30 (control unit 31). FIG. FIG. 11 is an explanatory diagram showing an example of a procedure for acquiring a second vector. FIG. 13 is a diagram showing an example of a recommended job offer presentation screen RD displayed on the recruiter terminal 20. FIG. 13 is a diagram showing an example of a job listing screen LD displayed on the recruiter terminal 20. This is an activity diagram showing the flow of information processing (recommended job offers and candidate presentation processing) performed by the recruitment support system 1.

The following describes embodiments of the present invention with reference to the drawings. The various features shown in the following embodiments can be combined with each other.

The program for implementing the software used in this embodiment may be provided as a non-transitory computer-readable recording medium, or may be provided so that it can be downloaded from an external server, or may be provided so that the program is started on an external computer and its functions are implemented on a client terminal (so-called cloud computing).

In this embodiment, a "unit" may also include, for example, a combination of hardware resources implemented by a circuit in the broad sense and software information processing that can be specifically realized by these hardware resources. In addition, this embodiment handles various types of information, which may be represented, for example, by physical values of signal values representing voltage and current, high and low signal values as a binary bit collection consisting of 0 or 1, or quantum superposition (so-called quantum bits), and communication and calculations may be performed on a circuit in the broad sense.

In the broad sense, a circuit is a circuit realized by at least appropriately combining a circuit, circuitry, a processor, and memory. In other words, it includes application specific integrated circuits (ASICs), programmable logic devices (e.g., simple programmable logic devices (SPLDs), complex programmable logic devices (CPLDs), and field programmable gate arrays (FPGAs)), etc.

1. Hardware Configuration This section describes the hardware configuration.

<Recruitment Support System 1>
1 is a configuration diagram showing a hiring support system 1. The hiring support system 1 comprises a communication line 2, a server device 10, a plurality of recruiter terminals 20, and a plurality of job seeker terminals 30. The server device 10, the recruiter terminals 20, and the job seeker terminals 30 are configured to be able to communicate with each other via the communication line 2. The connection between the server device 10, the recruiter terminals 20, and the job seeker terminals 30 may be wired or wireless.

The hiring support system 1 constitutes part of a recruitment and job search system used by multiple recruiters (first recruiter U1 and second recruiter U2) and multiple job seekers (first job seeker U3 and third job seeker U4). The hiring support system 1 mainly manages job seeker registration information and job postings. In one embodiment, the hiring support system 1 is made up of one or more devices or components. These components are described below.

<Server device 10>
2 is a block diagram showing a hardware configuration of the server device 10. The server device 10 includes a control unit 11, a storage unit 12, a communication unit 13, and a communication bus 14. The control unit 11, the storage unit 12, and the communication unit 13 are electrically connected to each other inside the server device 10 via the communication bus 14.

<Control unit 11>
The control unit 11 processes and controls the overall operation related to the server device 10. The control unit 11 is, for example, a central processing unit (CPU). The control unit 11 realizes various functions related to the server device 10 by reading out a predetermined program stored in the storage unit 12. That is, information processing by the software stored in the storage unit 12 can be specifically realized by the control unit 11, which is an example of hardware, and executed as each functional unit included in the control unit 11. These will be described in more detail in the next section. Note that the control unit 11 is not limited to being single, and may be implemented to have multiple control units 11 for each function. Also, a combination of these may be used.

<Storage unit 12>
The storage unit 12 stores various information defined by the above description. This can be implemented, for example, as a storage device such as a solid state drive (SSD) that stores various programs and the like related to the server device 10 executed by the control unit 11, or as a memory such as a random access memory (RAM) that stores temporarily required information (arguments, arrays, etc.) related to the program calculations. The storage unit 12 stores various programs, variables, etc. related to the server device 10 executed by the control unit 11.

<Communication unit 13>
The communication unit 13 is preferably a wired communication means such as USB, IEEE 1394, Thunderbolt (registered trademark), wired LAN network communication, etc., but may also include wireless LAN network communication, mobile communication such as 3G/LTE/5G, BLUETOOTH (registered trademark) communication, etc. as necessary. In other words, it is more preferable to implement it as a collection of multiple communication means. In other words, the server device 10 may communicate various information from the outside via the communication unit 13 and the network.

The server device 10 may be in an on-premise form or in a cloud form. As a server device 10 in a cloud form, the above-mentioned functions and processes may be provided in the form of, for example, SaaS (Software as a Service) or cloud computing.

<Employer terminal 20>
Fig. 3 is a block diagram showing the hardware configuration of the recruiter terminal 20 and the job seeker terminal 30. As shown in Fig. 3A, the recruiter terminal 20 comprises a control unit 21, a memory unit 22, a communication unit 23, an input unit 24, an output unit 25, and a communication bus 26. The control unit 21, the memory unit 22, the communication unit 23, the input unit 24, and the output unit 25 are electrically connected inside the recruiter terminal 20 via the communication bus 26. The explanation of the control unit 21, the memory unit 22, and the communication unit 23 is omitted here because it is the same as the explanation of each unit in the server device 10. Note that the recruiter terminal 20 may be a terminal operated by a human resources agency that communicates with job seekers on behalf of the recruiter.

<Input unit 24>
The input unit 24 accepts operation inputs made by the user. The operation inputs are transferred as command signals to the control unit 21 via the communication bus 26. The control unit 21 may execute predetermined control or calculations based on the transferred command signals as necessary. The input unit 24 may be included in the housing of the recruiter terminal 20, or may be attached externally. For example, the input unit 24 may be implemented as a touch panel integrated with the output unit 25. When the input unit 24 is implemented as a touch panel, the user can input tap operations, swipe operations, and the like to the input unit 24. As the input unit 24, a switch button, a mouse, a track pad, a QWERTY keyboard, and the like can be adopted instead of a touch panel.

<Output unit 25>
The output unit 25 displays a screen of a graphical user interface (GUI) that can be operated by the user. The output unit 25 may be included in the housing of the recruiter terminal 20, or may be attached externally. Specifically, the output unit 25 may be implemented as a display device such as a CRT display, a liquid crystal display, an organic EL display, or a plasma display. It is preferable that these display devices are implemented by selectively using them according to the type of recruiter terminal 20.

<Job Seeker Terminal 30>
3B , the job seeker terminal 30 includes a control unit 31, a memory unit 32, a communication unit 33, an input unit 34, an output unit 35, and a communication bus 36. The control unit 31, the memory unit 32, the communication unit 33, the input unit 34, and the output unit 35 are electrically connected inside the job seeker terminal 30 via the communication bus 36. Descriptions of the control unit 31, the memory unit 32, the communication unit 33, the input unit 34, and the output unit 35 will be omitted as they are similar to the descriptions of the respective units in the recruiter terminal 20.

2. Functional Configuration In this section, the functional configuration of the present embodiment will be described. Information processing by the software stored in the storage unit 12 is specifically realized by the control unit 11, which is an example of hardware, and can be executed as each functional unit included in the control unit 11 (a processor included in the recruitment support system 1).

Figure 4 is a block diagram showing the functions realized by the server device 10 (control unit 11), the employer terminal 20 (control unit 21), and the job seeker terminal 30 (control unit 31).

As shown in FIG. 4A, the server device 10 (control unit 11) includes a basic display control unit 111, a job offer extraction unit 112, a candidate extraction unit 113, a presentation unit 114, a scout document management unit 115, and an artificial intelligence unit 120. As shown in FIG. 4B, the recruiter terminal 20 (control unit 21) includes a display unit 211 and an operation acquisition unit 212. As shown in FIG. 4C, the job seeker terminal 30 (control unit 31) includes a display unit 311 and an operation acceptance unit 312.

<Basic display control unit 111>
The basic display control unit 111 is configured to display various information on the recruiter terminal 20 and the job seeker terminal 30. For example, the basic display control unit 111 displays a resume and curriculum vitae prepared by the job seeker, a job advertisement and scouting document prepared by the recruiter, etc. on the display unit 211 of the recruiter terminal 20 or the display unit 311 of the job seeker terminal 30.

Employers include organizations such as profit-making corporations (e.g., companies), non-profit organizations (e.g., cooperatives, foundations), and public organizations (e.g., local governments). Employers also include recruitment agencies that act as agents of organizations to mediate between job seekers and organizations. Recruitment agencies are also called headhunters or agents.

<Job Extraction Unit 112>
The job offer extraction unit 112 is configured to extract at least one recommended job offer from among the job offers registered in the database by the recruiter, based on the content of the job offer (job offer form data) and a contract expectation value calculated based on the first reference information. The contract expectation value is a numerical value indicating the likelihood of a contract, and it is estimated that the higher the contract expectation value of a job offer, the higher the likelihood of a contract.

A job posting, which is the content of a job offer, lists job information for multiple items such as the name of the position being advertised, the job content and working conditions (annual salary, job type, industry, work location, working style, work environment, etc.), application qualifications (skills), the desired personality, and selling points. A job posting may also include items such as the job title, headline, and information about the employer (company size (sales, number of employees, etc.), industry, etc.). Job postings are registered in a recruiter database stored in, for example, the memory unit 12. The recruiter database stores the registration information (organizational information) of the user, who is the employer, and the job posting created by the employer.

The first reference information includes a correlation between the content of the job offer and an expected value for closing, which is the probability of closing the job offer for that job offer. The first reference information is stored, for example, in the storage unit 12. The first reference information is an estimator constructed so that it can output an expected value for closing the job offer when a job posting is input. The first reference information may be, for example, a table, a function, a simple algorithm, or the like, which indicates a correlation between a feature amount (vector data) extracted from the job posting and the closing result. The correlation included in the first reference information is constructed, for example, by statistically analyzing data that records the closing results for actual job postings.

The first reference information may be an expectation value calculation model that has been trained to be able to input the content of a job offer and output the expected value of a successful contract. In this case, the job offer extraction unit 112 inputs the content of the job offer into the expected value calculation model of the artificial intelligence unit 120, and causes the expected value calculation model to output the expected value of a successful contract. This makes it possible to predict the expected value of a successful contract based on the results of successful contracts for a large number of job offers. The expected value calculation model outputs the expected value of a successful contract as a number between 0 and 1, for example.

The expected value calculation model is a learning model that learns from training job posting data and data on the success of job offers in the job posting (typically data on whether a job offer has been concluded and when it was concluded) as training data. The training job posting data includes master information (numeric values, or items selected from a set of pre-defined candidates) such as annual salary, job type, industry, work location, and skills, as well as text information consisting of sentences such as titles and detailed information.

The expected value calculation model is preferably trained using the contents of multiple job offers and whether or not these job offers have been concluded within a predetermined first period. In this way, for example, by learning whether or not job offers have been concluded in a period close to the present time, it is possible to output an expected value for conclusion that reflects the most recent tendency for conclusion. As a result, recommended job offers with a higher probability of conclusion can be presented to the user (the job offeror).

The first specified period is, for example, a period within six months from the time of learning. Data on job offers that have been concluded but that are outside the first specified period (for example, a contract concluded one year ago) is either used to learn the expected value calculation model as training data for "job offers without contracts" or is excluded from the training data for the expected value calculation model. For example, the expected value calculation model may be trained using data on multiple job offers without contracts and data on multiple job offers that were concluded within the first specified period.

In addition, only job offers whose period of time since registration is within a predetermined second designated period may be used as learning subjects (teaching data) for the expected value calculation model, regardless of whether the job offer has been concluded or not. In other words, job offers whose period of time since registration exceeds the second designated period may be excluded from learning subjects for the expected value calculation model. The second designated period is, for example, one year. The expected value calculation model is subjected to additional learning (updating of the learning model) based on data on newly registered job offers and newly concluded job offers at regular intervals (for example, once a day) by, for example, the artificial intelligence unit 120.

The expectation value calculation model may be a generative AI including a large-scale language model. In this case, the job extraction unit 112 inputs a job posting, inputs a prompt including an instruction to output a contract expectation value to the expectation value calculation model, and causes the expectation value calculation model to output the contract expectation value. In addition to the job posting and the instruction to output the contract expectation value, the job extraction unit 112 may also input a prompt into the expectation value calculation model that includes, for example, one or more job posting samples and one or more corresponding contract expectation value samples, as input and output samples.

<Another example of how to calculate expected value of contract>
The expected value of contract may be calculated based on expected value information regarding the occurrence rate or number of actions on a job posting for which the expected value of contract is to be calculated (hereinafter, the "target job posting"). That is, the job extraction unit 112 may be configured to generate expected value information based on the contents of the target job posting and the first reference information. The "action on the target job posting" includes employer actions, which are actions taken by employers based on the target job posting, and job seeker actions, which are actions taken by job seekers on the target job posting.

Employer actions include direct actions initiated by the target job posting (e.g., sending a scouting document based on the target job posting) and indirect actions initiated by another action (e.g., sending a scouting document and making a job offer for a job posting for which a job seeker has replied, or closing a job posting for which a scouting document has been sent). Job seeker actions also include direct actions initiated by the target job posting (e.g., applying for a job posting) and indirect actions initiated by another action (e.g., replying to a scouting document sent by the employer, accepting a job offer from the employer, or closing a job posting for which a job seeker has applied).

Specifically, the employer actions that the job extraction unit 112 refers to when calculating expected value information may include at least one of sending a scouting document to a job seeker based on the target job posting, the job seeker's reply to the scouting document, and the conclusion of a job offer with the job seeker. This makes it possible to obtain the expected value of each action that begins with the sending of a scouting document from the employer.

In addition, the job seeker actions that the job extraction unit 112 refers to when calculating the expected value information may include at least one of a job seeker's application for the target job posting and a job offer made to a job seeker who has applied. This makes it possible to obtain the expected value of each action that begins with a job seeker's application.

Furthermore, the job extraction unit 112 may generate information on expectations regarding passing a job interview (first interview, second interview, etc.) and receiving a job offer as an action based on the target job posting.

The first reference information used when calculating the expected value information includes a correlation between the content of the job offer and the occurrence or number of occurrences of an action in the job offer. The first reference information is an estimator constructed so that, using a target job posting as input, it is possible to output expected value information for the target job posting (the predicted occurrence rate or predicted number of occurrences of an action).

The first reference information may be an expected value information output model that has been trained to be capable of inputting the content of the job offer (target job posting) and outputting expected value information. The expected value information output model constitutes a part of the expected value calculation model. In this case, the job offer extraction unit 112 inputs the target job posting to the expected value information output model of the artificial intelligence unit 120, and causes the expected value information output model to output expected value information. This makes it possible to generate expected value information based on the occurrence or absence or number of actual actions in past job postings, by learning and modeling the tendency of job postings that are likely to result in actions such as contract (decision to hire the job seeker) based on a model that has learned information from the reference job posting, which is an actually used job posting, and thereby improving the accuracy of the predicted occurrence rate or number of actions.

The expected value information output model is a learning model that predicts whether or not an action will occur in a target job posting, or the number of times it will occur. The expected value information output model is a learning model that learns using reference job postings for learning and data on the occurrence rate or number of occurrences of corresponding actions as training data. A "reference job posting" is a job posting that is registered or has been registered in the job posting database, and that has actually posted a job posting in the past or is currently posting a job posting. In this case, the parameters of the expected value information output model that have been calculated or tuned by learning correspond to the correlation between the content of the job posting and whether or not an action will occur in the job posting, or the number of times it will occur.

The expected value information output model may be a generative AI including a large-scale language model. In this case, the job extraction unit 112 inputs the target job posting, inputs a prompt including an instruction to predict and output the occurrence rate or number of occurrences of the action of the target job posting to the expected value information output model, and causes the expected value information output model to output the occurrence rate or number of occurrences of the action. In addition to the target job posting and the instruction to predict and output the occurrence rate or number of occurrences of the action, the job extraction unit 112 may also input a prompt into the expected value information output model in which, for example, one or more samples of job postings and one or more samples of the occurrence rate or number of corresponding actions are inserted as input and output samples.

The job extraction unit 112 may generate the probability of a job offer being concluded through the scout document as the expected value information. This makes it possible to obtain the probability of a job offer being concluded for the target job posting. The probability of a job offer being concluded may be expressed as a probability such as a percentage, or may be acquired and expressed as a score. A job offer being concluded means that a job seeker accepts a job offer from a recruiter and the job seeker is decided to be employed for the target job posting, and may be called a job offer decision. The probability of a job offer being concluded may be rephrased as the ease of conclusion, the expectation of conclusion, the possibility of conclusion, the ease of decision, the expectation of decision, or the possibility of decision. In this case, the job extraction unit 112 uses a first expected value information output model that predicts the probability of a job offer being concluded for the target job posting as the expected value information output model. The first expected value information output model is a learning model that is trained to input the target job posting and output the probability of a job offer being concluded (i.e., the occurrence rate of the action of a job offer being concluded). The first expectation information output model is a learning model that is trained using a reference job posting for learning and corresponding data on whether or not a job posting has been concluded as training data. In this case, the parameters of the first expectation information output model calculated by learning correspond to the correlation between the contents of the job posting and the success rate in the job posting. The data on whether or not a job posting has been concluded is binarized data, for example, with "1" indicating a successful conclusion and "0" indicating no successful conclusion. The first expectation information output model trained using such data outputs the success probability as a number in the range from 0 to 1.

The first expected value information output model may be a generative AI including a large-scale language model. In this case, the job offer extraction unit 112 inputs the target job posting, inputs a prompt including an instruction to predict and output the probability of the job posting of the target job posting to the first expected value information output model, and causes the first expected value information output model to output the probability of the job posting. In addition to the instruction to predict and output the probability of the job posting and the target job posting, the job offer extraction unit 112 may input a prompt into the first expected value information output model, in which, for example, one or more samples of the job posting and one or more corresponding samples of the probability of the job posting are inserted as input and output samples. When the first expected value information output model is a generative AI, the job offer extraction unit 112 may cause the first expected value information output model to output text representing the probability of the job posting, such as "easy to conclude" or "difficult to conclude". The text to be output is selected based on the relationship between the threshold value and the probability of the job posting (whether or not the threshold value is exceeded). The threshold is set based on a statistical value, such as the average probability of closing a job offer.

The first expected value information output model may be learned using a reference job posting in which the number of scout documents transmitted based on the reference job posting is greater than a first threshold value determined in advance, and the presence or absence of a job offer in the reference job posting. That is, the first expected value information output model may be machine-learned using a reference job posting in which a certain number of scout documents have been transmitted or more, and data on the presence or absence of a job offer in the reference job posting as training data. The first expected value information output model may also be a generation AI including a large-scale language model that has been fine-tuned or otherwise trained using a reference job posting in which a certain number of scout documents have been transmitted or more, and data on the presence or absence of a job offer in the reference job posting. In order to reach a job offer, it is often necessary to transmit a certain number of scout documents, and a job posting in which the number of scout documents transmitted is small (or no scout documents have been transmitted) may have a low probability of a job offer being concluded. Therefore, by excluding reference job postings with a small number of scouting documents sent (or no scouting documents sent) from the learning data, the prediction accuracy of the first expectation information output model is improved.

The first threshold value for the number of scout documents sent for the reference job posting used in training the first expectation information output model is an arbitrary value, such as 5, 10, 20, etc. In other words, for example, only reference job postings for which 5, 10, or 20 or more scout documents have been sent are used in training the first expectation information output model.

The first expectation information output model may also weight the learning of each reference job posting according to the number of scout documents sent. For example, the first expectation information output model may learn using data on reference job postings that are weighted more heavily the more scout documents sent based on the reference job posting, and whether or not a job offer has been concluded in the reference job posting.

Furthermore, the first expected value information output model may be trained using reference job postings in which the number of scout documents sent based on the reference job posting is greater than a first predetermined threshold value, and the presence or absence of a job offer agreement in the reference job posting within a predetermined period from the sending of the scout document. That is, in the training of the first expected value information output model, a label of "job offer agreement" is attached to reference job postings in which a certain number of scout documents have been sent or more, in which a job offer agreement has been reached within a certain period from the sending of the scout document. Also, a label of "no job offer agreement" is attached to reference job postings in which a certain number of scout documents have been sent or more, in which there has been no job offer agreement, or if there has been a job offer agreement, it has only been after the passage of a certain period. Then, the reference job postings with these labels are used as training data. If no conditions are set for the period from the sending of the scout document to the job offer agreement, the number of job offers that are concluded increases over time, and the probability of a job offer agreement may become too high. Therefore, by setting a condition for the period from the sending of the scout document to the conclusion of a job offer in the labeling of the learning data, the number of reference job postings for which a job offer has been concluded can be prevented from continuing to increase, and appropriate learning data can be used. In addition, the conditions for conclusion of a job offer between a reference job posting registered in the database at a relatively old time and a reference job posting registered in the database at a relatively new time can be aligned. As a result, the prediction accuracy of the first expectation information output model is improved. The first expectation information output model may be trained using reference job postings that have been concluded within a predetermined period from the registration or publication of the job posting.

The period of time for which a job offer has been concluded for labeling the data for training the first expected value information output model as "job offer concluded" is any period, such as within 6 months, 12 months, or 18 months from the date of sending the scouting document. In other words, only reference job postings for which a job offer has been concluded within, for example, 6 months, 12 months, or 18 months from the date of sending the scouting document are used as data labeled as "job offer concluded."

When multiple deals have been concluded for one reference job posting, i.e., when multiple job seekers have been employed for one reference job posting, the first expectation information output model may treat the number of deals concluded per scouting document (i.e., a value normalized by dividing the number of scouting document transmissions by the number of job offers concluded) as the number of deals concluded for the scouting documents for that reference job posting. The number of scouting document transmissions normalized in this way is used in the decision to accept or reject the data for learning and in weighting as described above.

The conditions for the reference job posting to be used in the learning of the first expectation information output model may further include the period of time elapsed from the date of registration in the database or the date of publication to job seekers, and the period of time from the date of registration or publication to the date of sending the scout document. The publication date is the date of posting on the job posting/job search service, or the date of publication on the Internet. For example, in addition to satisfying the above-mentioned condition of the number of scout documents sent, the first expectation information output model may use only reference job postings that were registered between 18 months ago and 6 months ago from the present time and for which the scout document was sent within 6 months from the registration date or publication date as learning data. The balance between the number of job seekers and the number of job postings may differ depending on the time period, and the ease of closing a deal may differ. However, by limiting the time period when the scout document was sent, the variation in closing due to the difference in the balance between the number of job seekers and the number of job postings at that time period can be suppressed, and the lead time from the registration of the job posting to closing is also taken into account, improving the prediction accuracy of the first expectation information output model.

The job offer extraction unit 112 may generate the probability of a reply from a job seeker to a scout document as the expected value information. Compared to a contract, a reply from a job seeker to a scout document occurs in a short period from the registration of the job offer, so by obtaining the probability of a reply to the scout document, a score using new information more recent than the contract of a job offer can be obtained. In this case, the job offer extraction unit 112 uses a second expected value information output model that predicts the probability of a reply from a job seeker for a target job offer as the expected value information output model. The second expected value information output model is a learning model that uses the target job offer as an input and is trained to output the probability of a reply from a job seeker (i.e., the occurrence rate of the action of replying to a scout document). In other words, the second expected value information output model is a learning model that is trained using the reference job offer and the corresponding data on the presence or absence of a reply to the scout document as training data. In this case, the parameters of the second expected value information output model calculated by learning correspond to the correlation between the contents of the job offer and the probability of a reply from a job seeker. In addition, the data on the presence or absence of a reply to the scout document is binarized data, for example, with a reply being "1" and no reply being "0." The second expected value information output model trained using such data outputs the reply probability as a numerical value in the range from 0 to 1.

The second expected value information output model may be a generative AI including a large-scale language model. In this case, the job offer extraction unit 112 inputs the target job posting, inputs a prompt including an instruction to predict and output the probability of a reply to the scout document of the target job posting, into the second expected value information output model, and causes the second expected value information output model to output the probability of a reply to the scout document. In addition, the job offer extraction unit 112 may input a prompt into the second expected value information output model, in addition to the instruction to predict and output the probability of a reply to the scout document and the target job posting, as input and output samples, for example, a sample of one or more job postings and a sample of the probability of a reply to one or more corresponding scout documents. When the second expected value information output model is a generative AI, the job offer extraction unit 112 may cause the second expected value information output model to output text representing the probability of a reply, such as, for example, "easy to reply" or "hard to reply". The text to be output is selected based on the relationship between the threshold and the probability of a reply (whether or not it exceeds the threshold). The threshold is set based on a statistical value, such as the average probability of a reply.

The second expectation information output model may be trained using reference job postings in which the number of scout documents sent based on the reference job posting is greater than a second threshold value determined in advance, and the presence or absence of replies to the scout documents in the reference job posting. That is, the second expectation information output model may be machine-trained using reference job postings in which a certain number of scout documents have been sent or more, and data on the presence or absence of replies to the scout documents in the reference job posting as training data. The second expectation information output model may also be a generation AI including a large-scale language model that has been trained by fine tuning or the like using reference job postings in which a certain number of scout documents have been sent or more, and data on the presence or absence of replies to the scout documents in the reference job posting. As a result, reference job postings in which a small number of scout documents have been sent (or no scout documents have been sent) are excluded from the training data, improving the prediction accuracy of the second expectation information output model.

The second threshold for the number of scout documents sent for a reference job posting used in the training of the second expectation information output model is a number (e.g., 5) smaller than the first threshold in the first expectation information output model. Similarly to the first expectation information output model, the second expectation information output model may weight the training of each reference job posting according to the number of scout documents sent. For example, the second expectation information output model may train using data on reference job postings that are weighted more heavily the greater the number of scout documents sent based on the reference job posting, and the presence or absence of a reply to the scout document in the reference job posting.

Furthermore, the second expectation information output model may be trained using reference job postings in which the number of scout documents sent based on the reference job posting is greater than a predetermined threshold, and the presence or absence of replies to the scout documents in the reference job postings within a predetermined period from the sending of the scout documents. In other words, in the training of the second expectation information output model, among the reference job postings in which a certain number of scout documents have been sent or more, those for which a reply was received within a certain period from the sending of the scout document are labeled as "reply received," and those for which there was no reply, or if there was a reply, it was only a reply after a certain period has passed are labeled as "no reply," and are used as training data. In this way, by setting a condition for the period from the sending of the scout document to the reply in the labeling of the training data, the number of reference job postings for which replies were generated is prevented from continuing to increase, thereby improving the prediction accuracy of the second expectation information output model.

The reply period for labeling the training data for the second expectation information output model with a "reply received" label is, for example, within 14 days from the sending of the scout document, which is shorter than the period for job contract completion (for example, within 6 months) for labeling the training data for the first expectation information output model with a "job contract completed" label. In other words, only reference job postings that have received a reply within 14 days from the sending of the scout document are used as data labeled with a "reply received".

Conditions for the reference job posting used in training the second expectation information output model may further include the period of time that has elapsed since the date of registration in the database or the date of publication to job seekers, and the period of time from the date of registration or publication to the sending of the scouting document. For example, in addition to satisfying the above-mentioned number of scouting documents sent, the second expectation information output model may only use reference job postings that were registered between 12 months and 14 days prior to the present time and for which a scouting document was sent within one year of the date of registration or publication as training data. This improves the accuracy of the second expectation information output model, as it is possible to predict the probability using a more recent reference job posting than the first expectation information output model.

The job offer extraction unit 112 may generate a predicted value of the number of scout document transmissions as the expected value information. If the number of candidates who match the contents of the job offer is small and the number of scout documents that the employer can send is small, it may be difficult to recruit personnel even if the probability of job offer contract or the probability of reply to the scout document is high. By generating a predicted value of the number of scout document transmissions, the expected value of the number of scout document transmissions based on the target job offer can be obtained as the expected value of contract. As a result, it is possible to evaluate the job offer based on the predicted number of scout document transmissions. In this case, the job offer extraction unit 112 uses a third expected value information output model that predicts the number of scout document transmissions for the target job offer as the expected value information output model. The third expected value information output model is a learning model that is trained to input the target job offer and output the expected value of the number of scout document transmissions (i.e., the number of occurrences of the action of sending to the scout document). In other words, the third expected value information output model is a learning model that is trained using the reference job posting and the corresponding data on the number of scout document transmissions as training data. In this case, the parameters of the third expected value information output model calculated by learning correspond to the correlation between the content of the job posting and the number of scout document transmissions.

The third expected value information output model may be a generative AI including a large-scale language model. In this case, the job extraction unit 112 inputs the target job posting, inputs a prompt including an instruction to predict and output the number of scout document transmissions based on the target job posting, and causes the third expected value information output model to output the expected value of the number of scout document transmissions. In addition, the job extraction unit 112 may input a prompt into the third expected value information output model, in which, for example, one or more samples of job postings and one or more corresponding samples of the expected value of the number of scout document transmissions are inserted as input and output samples, in addition to the instruction to predict and output the number of scout document transmissions and the target job posting. When the third expected value information output model is a generative AI, the job extraction unit 112 may cause the third expected value information output model to output text representing the expected value of the number of scout document transmissions, such as "easy to send" and "difficult to send". The text to be output is selected based on the relationship between the threshold value and the expected value of the number of transmissions (whether or not the threshold value is exceeded). The threshold is set based on a statistical value, such as the average number of transmissions, for example.

The third expectation information output model may be trained using a reference job posting that is within a predetermined period of time from the date of registration in the database or the date of publication to job seekers, and the number of scout documents sent in the reference job posting. In other words, the third expectation information output model may be machine-trained using a reference job posting (including those with zero scout document transmissions) that has not yet passed a certain period of time since the date of registration or publication, and data on the number of scout documents sent in the reference job posting as training data. The third expectation information output model may also be a generation AI including a large-scale language model that has been trained by fine tuning, etc., using a reference job posting that has not yet passed a certain period of time from the date of registration or publication, and data on the number of scout documents sent in the reference job posting. This makes it possible to suppress variations in the number of scout documents sent due to differences in the balance between the number of job postings and job seekers at the time of registration of the job posting.

Furthermore, the third expectation information output model may be trained using a reference job posting and the number of scout document transmissions in the reference job posting within a predetermined period from the registration date or publication date of the reference job posting. That is, in training the third expectation information output model, the number of scout documents transmitted within a certain period from the registration date or publication date of the reference job posting is labeled as the "number of scout document transmissions" and used as training data. Therefore, scout documents transmitted after a certain period from the registration date or publication date are not included in the "number of scout document transmissions." In this way, by setting a condition for the period from the registration date or publication date to the transmission of the scout document in the labeling of the training data, the number of scout document transmissions in each reference job posting is prevented from continuing to increase, improving the prediction accuracy of the third expectation information output model.

In the data for learning the third expected value information output model, the transmission period of the scouting documents to be counted as the "number of scouting documents sent" is, for example, within six months from the date of registration or publication of the reference job posting. In other words, only scouting documents sent within six months from the date of registration or publication of the reference job posting are counted as the "number of scouting documents sent."

The job extraction unit 112 calculates the expected value of a deal from the expected value information. Specifically, the job extraction unit 112 generates first expected value information, which is the probability of a job offer being concluded, second expected value information, which is the probability of a reply from a job seeker, and third expected value information, which is a predicted value of the number of scout documents to be sent, and calculates the expected value of a deal for the job posting based on the first expected value information, the second expected value information, and the third expected value information. This makes it possible to assign an overall score (expected value of a deal) to the target job posting that combines the three expected value information.

For example, the job offer extraction unit 112 may calculate the average of the weighted values of the first expectation information, the second expectation information, and the third expectation information as the expected value of the conclusion of the job posting. The weighting of the first expectation information may be smaller than the weighting of the second expectation information and the third expectation information. This allows the calculation of the expected value of the conclusion to be weighted on the second expectation information (scout response rate) and the third expectation information (number of scout transmissions), which are more recent than the first expectation information (job offer conclusion rate) and also have a correlation with the conclusion of the job posting. As a result, a highly reliable expected value of the conclusion can be assigned to the target job posting. The job offer extraction unit 112 may also calculate the expected value of the conclusion from the first expectation information, the second expectation information, and the third expectation information by other methods, without relying on weighting.

The job extraction unit 112 may normalize the expected value of contract to a value between 0 and 1. Furthermore, the job extraction unit 112 may rank the target job listings with labels such as "S", "A", "B", "none", etc. according to the distribution of the normalized expected value of contract. For example, "S" is assigned to target job listings that have an expected value of contract that is in the top xx% or more of the distribution of expected values of contract.

The distribution of expected success values used for ranking is determined based on the evaluation criteria of the target job posting set by the employer. For example, if job postings handled by individuals or departments are to be evaluated, the target job postings are ranked using the distribution of expected success values for job postings handled by individuals or departments. Also, for example, if job postings handled by an organization are to be evaluated, the target job postings are ranked using the distribution of expected success values for job postings within the same organization. Furthermore, for example, if one wishes to check the evaluation of job postings issued by an organization in the world as a whole or within a given industry, the target job postings are ranked using the distribution of expected success values for all job postings, including those of other organizations.

The job extraction unit 112 may classify each of the first expectation information, the second expectation information, and the third expectation information into a level, such as "high" or "low," and generate a closing expectation value that combines the level of the first expectation information, the level of the second expectation information, and the level of the third expectation information.

The job extraction unit 112 may also input the target job posting to the expectation information output model and cause the expectation information output model to output the expected value of the contract (for example, the average value of the weighted values of the first expectation information, the second expectation information, and the third expectation information). In this case, the job extraction unit 112 uses the integrated expected value information output model as the expected value information output model. The integrated expected value information output model is a learning model that is trained to input the target job posting and output the expected value of the contract. In other words, the integrated expected value information output model is a learning model that is trained using the reference job posting and the corresponding data of the expected value of the contract as teacher data. The integrated expected value information output model may be a generative AI including a large-scale language model. In this case, the job extraction unit 112 inputs the target job posting, inputs a prompt including an instruction to calculate and output the expected value of the contract of the target job posting to the integrated expected value information output model, and causes the integrated expected value information output model to output the expected value of the contract. Furthermore, the job extraction unit 112 may input, as input and output samples, a prompt into the integrated expected value information output model, in addition to the calculation and output instructions for the expected value of a contract and the target job posting, for example, one or more sample job postings and one or more corresponding samples of the expected value of a contract. If the integrated expected value information output model is a generative AI, the job extraction unit 112 may cause the integrated expected value information output model to output text indicating the probability of a contract, such as "likely to be concluded" or "unlikely to be concluded".

The job extraction unit 112 may generate a predicted value of the number of applications from job seekers as the expected value information. This makes it possible to obtain the expected value of the number of applications from job seekers based on the target job posting as the expected value of contracts. In this case, the job extraction unit 112 uses the fourth expected value information output model as the expected value information output model. The fourth expected value information output model is a learning model that takes the target job posting as input and is trained to output the expected value of the number of applications from job seekers (i.e., the number of occurrences of the action of applying from a job seeker). In other words, the fourth expected value information output model is a learning model that is trained using the reference job posting and the corresponding data on the number of applications from job seekers as training data.

The fourth expectation information output model may be a generative AI including a large-scale language model. In this case, the job offer extraction unit 112 inputs the target job posting, inputs a prompt including an instruction to predict and output the number of applications from job seekers based on the target job posting, to the fourth expectation information output model, and causes the fourth expectation information output model to output the expected value of the number of applications from job seekers. In addition to the instruction to predict and output the number of applications from job seekers and the target job posting, the job offer extraction unit 112 may input a prompt into the fourth expectation information output model, in which, for example, a sample of one or more job postings and a sample of the expected value of the number of applications from one or more job seekers corresponding thereto are inserted as input and output samples. When the fourth expectation information output model is a generative AI, the job offer extraction unit 112 may cause the fourth expectation information output model to output text representing the expected value of the number of applications, such as, for example, "easy to apply" or "hard to apply". The text to be output is selected based on the relationship between the threshold value and the expected value of the number of applications (whether or not the threshold value is exceeded). The threshold value is set based on a statistical value such as the average value of the number of applications, for example.

The job extraction unit 112 may generate the probability of a job offer being concluded through a voluntary application from a job seeker as the expected value information. This makes it possible to obtain the probability of a job offer being concluded in the target job posting as the expected value of the offer. In this case, the job extraction unit 112 uses the fifth expected value information output model as the expected value information output model. The fifth expected value information output model is a learning model that is trained to take the target job posting as input and output the probability of a job offer being concluded (i.e., the occurrence rate of the action of a job offer being concluded). In other words, the fifth expected value information output model is a learning model that is trained using the reference job posting and the corresponding data on whether or not a job offer has been concluded as training data.

The fifth expected value information output model may be a generative AI including a large-scale language model. In this case, the job extraction unit 112 inputs the target job posting, inputs a prompt including an instruction to predict and output the probability of the job posting of the target job posting to the fifth expected value information output model, and causes the fifth expected value information output model to output the probability of the job posting. In addition to the instruction to predict and output the probability of the job posting and the target job posting, the job extraction unit 112 may input a prompt into the fifth expected value information output model, in which, for example, one or more samples of the job posting and one or more corresponding samples of the probability of the job posting are inserted as input and output samples. When the fifth expected value information output model is a generative AI, similar to the first expected value information output model, the job extraction unit 112 may cause the fifth expected value information output model to output text representing the probability of the job posting, such as "easy to conclude" or "difficult to conclude".

The fifth expectation information output model may be trained using reference job postings that have received more than a third threshold number of applications from job seekers and whether or not a job offer has been concluded in the reference job posting. In other words, the fifth expectation information output model may be machine-trained using reference job postings that have received a certain number of applications and data on whether or not a job offer has been concluded in the reference job posting as training data. The fifth expectation information output model may also be a generative AI that includes a large-scale language model that has been trained by fine tuning or the like using reference job postings that have received more than a certain number of applications and data on whether or not a job offer has been concluded in the reference job posting. This allows reference job postings that have received a small number of applications (or no applications) to be excluded from the training data, thereby improving the prediction accuracy of the fifth expectation information output model.

The third threshold value for the number of applications for the reference job posting used in training the fifth expectation information output model is, for example, 100. In other words, only reference job postings with 100 or more applications are used in training the fifth expectation information output model.

Specifically, the feature quantities of a job posting are used as input to the expected value calculation model or the expected value information output model. The feature quantities of a job posting are data derived from the contents of the job posting, regardless of the format of the job posting. As the feature quantities, for example, feature vectors obtained by converting the contents of the target job posting into vectors expressing its features are used. Here, multiple feature vectors may be obtained from one target job posting, and multiple feature vectors may be used as feature quantities to be input to the expected value calculation model or the expected value information output model. Note that the feature quantities of a job posting may be extracted by a method other than vectorization.

Specifically, the job extraction unit 112 inputs feature quantities including a first vector obtained by vectorizing the text included in the job posting to the expectation calculation model or the expectation information output model, and causes the expectation calculation model or the expectation information output model to output the contract expectation or the expectation information. This eliminates the effects of the job posting format (type of items, order of items, etc.) and spelling variations, and allows the contract expectation or the expectation information based on text information such as the text included in the job posting to be output. This improves the accuracy of the contract expectation or the expectation information.

The job extraction unit 112 may also input a feature quantity including a second vector obtained by vectorizing the attribute information included in the job posting to the expectation value calculation model or the expectation value information output model, and cause the expectation value calculation model or the expectation value information output model to output the contract expectation value or the expectation value information. This improves the accuracy of the contract expectation value or the expectation value information based on the category information (annual salary, job type, industry, work location, etc.) included in the job posting.

Furthermore, the job extraction unit 112 may input a feature quantity including both the first vector and the second vector to an expectation value calculation model or an expectation value information output model, and cause the expectation value calculation model or the expectation value information output model to output a contract expectation value or expectation value information.

The expected value calculation model or expected value information output model is trained to input features including at least one (preferably both) of a first vector obtained by vectorizing the text included in the job posting and a second vector obtained by vectorizing the attribute information included in the job posting, and to output the expected value of the contract or expected value information.

The job extraction unit 112 converts the sentences included in the job posting into a first vector, for example, in the following procedure. First, the job extraction unit 112 performs morphological analysis on the text data included in the job posting data and divides the sentences into words. Furthermore, the divided words are filtered to remove stop words (functional words such as particles and auxiliary verbs) and extract only nouns. Stop words are determined, for example, based on dictionary definitions, frequency of occurrence, etc. Next, the job extraction unit 112 performs word normalization (absorption of spelling variations) on the extracted nouns. Word normalization includes procedures such as standardizing character sets, replacing numbers, and standardizing words using a dictionary. Standardizing character sets includes standardizing uppercase letters to lowercase letters, standardizing half-width kana to full-width kana, etc. Number replacement is replacing numbers included in words with representative symbols representing numbers (for example, "0").

Then, the job extraction unit 112 uses the text of the job posting and the words that have been processed as described above to obtain a first vector using any vectorization method, such as the TF-IDF method.

The TF-IDF method is a method of quantifying importance using the frequency of occurrence of words in a sentence, and the job extraction unit 112 calculates the number of times each word contained in the sentence of a job posting appears in that job posting. The calculation of the number of times of occurrence is performed, for example, using a sentence analysis model such as Bag of Words. Next, the job extraction unit 112 converts the number of times a word appears in one job posting to the term frequency of the word (tf), and further calculates the inverse document frequency (idf), which is the rarity of the frequency of occurrence of each word, in all job postings from which the data was obtained, based on the frequency of occurrence of the word (tf) in each job posting. Finally, the job extraction unit 112 calculates the importance (tf-idf) of each word in each job posting from the occurrence frequency (tf) and the inverse document frequency (idf), and generates a first vector whose components are the importance of each word. The dimension of the first vector is the number of words whose importance is calculated.

Note that in addition to the above-mentioned TF-IDF method, the LSI method or the LDA method may be used to quantify each word. Furthermore, the job extraction unit 112 may generate a first vector from the text of the job posting using a model that performs vectorization based on distributed representation of words, such as Word2vec or BERT. Note that the TF-IDF method has the advantage of being strong in keyword processing and light in processing. Since keywords that appear in job postings have characteristics, it is preferable to use the TF-IDF method. Furthermore, the TF-IDF method, for example, reduces the weight of words (such as "company" and "department") that appear in many job postings regardless of whether the job posting has been concluded, and therefore the feature amount of the job posting can be suitably determined.

The job extraction unit 112 converts the attribute information included in the job posting into a second vector, for example, by the following procedure. The job extraction unit 112 converts the attribute information, which is numerical data or categorical (qualitative) data, into a vector, for example, by a method such as One-Hot Encoding. In One-Hot Encoding, the job extraction unit 112 converts the attribute information, which is numerical data or categorical (qualitative) data, into a One-Hot vector, each component of which is 0 or 1. The job extraction unit 112 generates the second vector by combining the One-Hot vectors of each piece of attribute information. "Attribute information" is information that is input by selecting from predetermined categories and values, for example, gender, age, annual income, industry, etc.

5 is an explanatory diagram showing an example of the procedure for obtaining the second vector. For example, as shown in FIG. 5A, the job offer extraction unit 112 converts the industries of each job posting shown in the table on the left into the One-Hot vector shown in the table on the right. In the One-Hot vector in FIG. 5A, if the industry corresponds to the industry assigned to each component, the component is "1", and if not, the component is "0". If the industry of the job posting corresponds to multiple industries of the components of the One-Hot vector, the multiple components are "1". Also, as shown in FIG. 5B, the job offer extraction unit 112 converts the annual income of each job posting shown in the table on the left into the One-Hot vector shown in the table on the right. In the One-Hot vector in FIG. 5B, the annual income is divided into multiple annual income bands in a certain interval, and if it corresponds to the annual income band assigned to each component, the component is "1", and if it does not correspond, the component is "0". If the annual salary listed in the job posting falls within multiple income bands in the One-Hot vector, multiple components will be "1".

The job extraction unit 112 may vectorize the job posting using a vectorization model, which is a generative AI including a large-scale language model. In this case, the job extraction unit 112 inputs the job posting, inputs a prompt including an instruction to vectorize and output the job posting to the vectorization model, and causes the vectorization model to output the first vector and/or the second vector. In addition to the job posting and the instruction to vectorize and output the job posting, the job extraction unit 112 may also input, to the vectorization model, a prompt in which, for example, one or more job posting samples and one or more corresponding first vector and/or second vector samples are inserted.

The job offer extraction unit 112 may extract job offers with a closing expectation value equal to or greater than a predetermined threshold as recommended job offers. This makes it possible to extract recommended job offers on the condition that the possibility of closing is at least a certain level. Therefore, job offers with a high possibility of closing can be presented preferentially to the user (employer). For example, if the closing expectation value is normalized to a value between 0 and 1, the job offer extraction unit 112 extracts job offers with a closing expectation value of 0.8 or greater as recommended job offers.

The threshold value for expected closing value may also be defined as a percentage (top percent) of the highest expected closing value for a specific job population. The "specific job population" is composed of all valid (i.e., currently open) job openings registered in the job database, or all valid job openings registered by user employers. For example, the job extraction unit 112 may extract job openings with expected closing values within the top 30% as recommended jobs.

Furthermore, the job extraction unit 112 may prepare multiple thresholds for the percentage from the top. For example, the job extraction unit 112 may first extract job offers whose expected value of success is equal to or greater than a first threshold (e.g., the top 30%) as recommended job offers, and if the number of extracted recommended job offers does not meet a predetermined reference number, extract job offers whose expected value of success is equal to or greater than a second threshold (e.g., the top 50%) as recommended job offers. If there are no job offers whose expected value of success is equal to or greater than the second threshold, the job extraction unit 112 determines that there are no recommended job offers.

The job extraction unit 112 may also recommend jobs that have been given a rank of a certain level or higher based on the distribution of expected contract values as described above.

The job offer extraction unit 112 may extract recommended job offers from among the job offers registered in the database by the employer, based on the expected value of the job offer and the history of sending scout documents based on the job offer to job seekers. This makes it possible to extract job offers with even lower utilization rates (low frequency of scouting or no scouting) from among job offers with a high probability of success, and present them to the user, the employer. As a result, in accordance with the demand from the job seeker, it is possible to encourage the employer to operate (search for job seekers, send scout documents, etc.) job offers that are not being used much but have a high probability of success, and to carry out recruitment activities efficiently.

Specifically, the job extraction unit 112 may extract recommended job offers based on a job offer score calculated for each job offer. The job offer score increases according to the magnitude of the expected value of closing and its weighting, and decreases according to the magnitude of the number of scout documents sent in a predetermined evaluation period and its weighting. The weighting of the expected value of closing is greater than the weighting of the number of sendings. This makes it possible to extract recommended job offers that place emphasis on the possibility of closing while also taking into account the occupancy rate of the job offer. Therefore, compared to when emphasis is placed on low occupancy rate, the possibility of closing can be increased when an employer carries out recruiting activities such as sending scout documents for recommended job offers.

For example, the job extraction unit 112 calculates a contract score by multiplying the contract expectation value, normalized to a value between 0 and 1, by a first weighting coefficient, and an operation score by multiplying the number of scout document transmissions during the evaluation period, normalized to a value between 0 and 1, by a second weighting coefficient, and determines the job score as a value obtained by subtracting the operation score from the contract score. Here, the first weighting coefficient is greater than the second weighting coefficient. The evaluation period can be, for example, within six months or within a year from the present time. The normalized number of scout document transmissions can be obtained, for example, by setting the number of transmissions of the job with the largest number of scout document transmissions during the evaluation period among the jobs registered by the user (employer) as a reference value, and dividing the number of scout document transmissions of each job in the evaluation period by the reference value.

Furthermore, the job offer extraction unit 112 may count the number of times a scout document is sent using a weighting that changes depending on the time of year. For example, the job offer extraction unit 112 may count the number of times a scout document is sent so that the further away the date and time of sending the scout document is from the present time (the less recently the job offer is in operation), the smaller the number of times the scout document is counted (for example, a scout document sent more than one month ago may be counted as 0.5 times instead of 1 time).

The job extraction unit 112 may also determine the operation score as the inverse of the number of scout document transmissions during the evaluation period, normalized to a value between 0 and 1, and determine the job score as the operation score multiplied by the expected value for success. In this case, the operation score will be closer to 0 the more transmissions there are (the more active the job is).

Instead of the number of scout documents sent, the recruitment score may be calculated using a recruitment operation parameter other than the number of scout documents sent from recruiters to job seekers, such as the number of applications from job seekers for a job, the number of views of a job by job seekers, or the number of registrations of a job in a job bookmark (favorite) list by job seekers. In other words, the recruitment score may be a score that increases according to the magnitude of the expected value of contract and its weighting, and decreases according to the magnitude of the recruitment operation parameter in a predetermined evaluation period and its weighting. The weighting of the expected value of contract is greater than the weighting of the recruitment operation parameter.

Furthermore, the job score may be a score that increases according to the magnitude of the expected value of closing and its weighting, and also increases according to the magnitude of the job operation parameter during a predetermined evaluation period and its weighting. This makes it possible to recommend job openings with a high expected value of closing from among job openings with high operating rates, thereby efficiently identifying job openings that should be focused on.

The job extraction unit 112 extracts, for example, job offers whose job offer score calculated from the contract score and operation score is equal to or exceeds a predetermined threshold as recommended job offers.

The job extraction unit 112 may calculate, for each job, a first job score and a second job score in which at least one of the expected value for closing and the number of scout documents sent during the evaluation period is weighted differently, and extract, as recommended job offers, at least one first recommended job offer extracted based on the first job score and at least one second recommended job offer extracted based on the second job score. This makes it possible to extract, for example, a first recommended job offer that emphasizes the possibility of closing and a second recommended job offer that emphasizes utilization rate more than the first recommended job offer. Therefore, recommended job offers can be presented to the user, the employer, from multiple perspectives, and appropriate job offers can be stably proposed, which leads to job offer operation (implementation of recruiting activities), such as sending scout documents.

The first and second job scores may differ in both the first weighting coefficient (weighting for expected value of contract) and the second weighting coefficient (weighting for the number of scout document transmissions) used in the calculation. Alternatively, only one of them may differ. For example, the first weighting coefficient of the first job score may be the same as the first weighting coefficient of the second job score, and the second weighting coefficient of the second job score may be greater than the second weighting coefficient of the first job score. In this case, the second job score is a score that places more importance on the utilization rate than the first job score. In addition, either the second weighting coefficient of the first job score or the second weighting coefficient of the second job score may be zero. In other words, one of the first job score and the second job score may be a score that does not take into account the utilization rate (the number of scout document transmissions).

The job extraction unit 112, for example, extracts at least one job offer whose first job offer score is equal to or greater than a predetermined threshold as a first recommended job offer, and extracts at least one job offer whose second job offer score is equal to or greater than a predetermined threshold as a second recommended job offer.

The job extraction unit 112 may further calculate a third job score for each of the first job score and the second job score, in which at least one of the first weighting coefficient or the second weighting coefficient is different, and extract a third recommended job based on the third job score.

The job extraction unit 112 extracts recommended job offers (calculates job offer scores) for each specified period. The job offer extraction unit 112 extracts recommended job offers at intervals of, for example, one day. The first reference information (expectation value calculation model) may be updated in accordance with the extraction of recommended job offers by the job offer extraction unit 112. In other words, the first reference information may be updated using the latest data (newly acquired information on job offers or job seekers) for each period in which the extraction of recommended job offers is performed.

<Candidate Extraction Unit 113>
The candidate extraction unit 113 is configured to extract at least one candidate for a recommended job offer from among job seekers registered in the database, based on a job seeker matching index calculated based on the content of the recommended job offer (job posting data) and the second reference information. The job seeker matching index is a numerical value indicating the degree of match (matching degree) of the attributes of a job seeker to the content of the recommended job offer, and it is presumed that the larger the job seeker matching index, the higher the suitability (probability of passing the screening) for the recommended job offer.

The second reference information includes a correlation between the content of the job offer and a job seeker matching index indicating the degree of suitability of the job seeker for the job offer. The second reference information is stored, for example, in the storage unit 12. The second reference information is an estimator constructed so as to be able to use a job posting as input and output a job seeker matching index for the job posting for each job seeker registered in the database. The second reference information may be, for example, a table, a function, a simple algorithm, or the like, indicating a correlation between the feature amount (vector data) extracted from the job posting, the feature amount (vector data) of the job seeker's registered information, and the job seeker matching index for the job posting. The correlation included in the second reference information is constructed, for example, by statistically analyzing data on the feature amount of the job posting, the feature amount of the job seeker, and the job seeker matching index that are associated with each other.

Vectorization of the job seeker's registered information is performed in the same manner as the vectorization of the job posting described above. Specifically, the candidate extraction unit 113 obtains a first vector obtained by vectorizing text such as the job description contained in the job seeker's registered information, and a second vector obtained by vectorizing attribute information such as experienced job type, experienced industry, desired job type, desired industry, and desired annual salary, by using a method such as a vectorization model, which is a generative AI including a large-scale language model.

The job seeker's registration information includes the job seeker's resume, curriculum vitae, and other profile information. A "resume" is a document that mainly describes the job seeker's profile, current situation, educational background, employment history, desired working conditions, etc., while a "curriculum vitae," also known as a résumé, is a document in which the job seeker conveys to the employer the job seeker's work history, experience, skills, qualifications, etc. related to his or her past work. The registration information may also include the job seeker's desired conditions (desired industry, desired job type, etc.).

The candidate extraction unit 113 calculates a job seeker matching index for job seekers who are registered in the job seeker database and who can receive a scout document from one recommended job posting (who can apply for the recommended job posting). Specifically, the candidate extraction unit 113 calculates a job seeker matching index for each job seeker using the correlation of reference job seekers (job seeker matching index for job postings) included in the second reference information. The reference job seekers are other job seekers with the same or similar attributes included in the registration information registered in the job seeker database, or other job seekers with similar behavioral history regarding job postings.

Examples of "attributes included in the registered information" include items such as job type, industry, skills, and qualifications. Similarity of attributes in the registered information is determined based on "first similarity judgment information" (information defining the range of similarity of attributes) included in the second reference information. The first similarity judgment information may define at least one similar keyword for a keyword indicating an attribute, or may define a similarity criterion for features obtained by vectorizing a category, keyword, or text indicating an attribute. The similarity criterion for features is a threshold for the difference (vector distance) between the features of two attributes, and for example, attributes whose difference in features is less than the threshold (or whose cosine similarity is equal to or greater than the threshold) are determined to be similar to each other.

The first similarity judgment information may also include information that groups similar attributes in advance. In this case, attributes that belong to the same group are judged to be similar to each other. For example, in the first similarity judgment information, "public relations," "marketing," "CRM," etc. are predefined as similar attributes (job types), and bundled into a group (group) named "marketing." The candidate extraction unit 113 judges job types that belong to the same group to be similar job types.

The closeness of the behavioral histories of job seekers is determined based on the "second similarity determination information" (information defining the similarity range of the behavioral histories) included in the second reference information. The second similarity determination information defines a threshold for similarity determination for the number of common job offers among the job offers that two job seekers have viewed, applied for, registered in a bookmark list, and replied to a scouting document sent based on a job offer, for example. In other words, two job seekers who have viewed a number of common job offers equal to or greater than the threshold are determined to have close (similar) behavioral histories based on the second similarity determination information. The second similarity determination information may define a criterion for similarity determination based on a combination of multiple types of behavioral histories (for example, a similarity determination using the number of common job offers among the job offers applied for and the number of common job offers among the job offers replied to the scouting document). As such a criterion, for example, a judgment formula that compares a value obtained by adding weights for the number of common job offers for each behavioral history with a threshold is used.

The second reference information may be a matching index calculation model that has been trained to be able to input the content of a job offer and output a job seeker matching index for each job seeker. In this case, the candidate extraction unit 113 inputs the content of the recommended job offer into the matching index calculation model of the artificial intelligence unit 120 and causes the matching index calculation model to output a job seeker matching index. This makes it possible to calculate a job seeker matching index based on the content of a large number of job offers and information on job seekers. The matching index calculation model outputs the job seeker matching index as a numerical value between 0 and 1, for example.

The matching index calculation model is a learning model that learns from training data on job postings, training data on job seekers (such as registration information and job-related action history), and/or data on the job seeker matching index of the job seeker for the job posting.

The matching index calculation model may be a learning model that learns the relationship between the contents of multiple job offers (job postings) and job seekers who have a history of actions related to the job offers. In this case, the matching index calculation model outputs a job seeker matching index as the degree of similarity between the input job offer and a job seeker who has a history of actions related to a similar job offer. "Action related to a job offer" refers to an action from a job seeker to a job seeker based on the job offer. This allows the matching index calculation model to calculate a job seeker matching index that reflects the likelihood of an action occurring based on the input recommended job offer.

The "job-related action" may be receiving a scouting document or passing a screening process. This makes it possible to extract as candidates job seekers who have received a scouting document from a job offer similar in content to the recommended job offer, or who have passed a screening process (document screening, interview screening, etc.).

The matching index calculation model may be a generative AI including a large-scale language model. In this case, the candidate extraction unit 113 inputs the job posting of the recommended job and information of the job seeker for which the index is to be calculated (registration information, history of actions related to the job, etc.), inputs a prompt including an instruction to output a job seeker matching index to the matching index calculation model, and causes the matching index calculation model to output the job seeker matching index. In addition, the candidate extraction unit 113 may input a prompt into the matching index calculation model that includes, as input and output samples, an instruction to output the job seeker matching index, the job posting, and the job seeker information, as well as one or more samples of the job posting and job seeker information and one or more corresponding samples of the job seeker matching index.

The candidate extraction unit 113 may extract as candidates job seekers whose job seeker matching index is equal to or greater than a predetermined threshold. This makes it possible to extract candidates on the condition that the degree of suitability for the recommended job offer is equal to or greater than a certain level. Therefore, job seekers who are highly suitable for the recommended job offer can be preferentially presented to the user, the employer. For example, if the job seeker matching index is normalized to a value between 0 and 1, the candidate extraction unit 113 extracts as candidates job seekers whose job seeker matching index is equal to or greater than 0.8. The threshold may also be defined as a percentage (top percent) of the multiple job seekers whose job seeker matching index is calculated. For example, the candidate extraction unit 113 may extract as candidates job seekers whose job seeker matching index is within the top 30%.

The candidate extraction unit 113 may extract candidates from among the job seekers registered in the database based on the job seeker matching index of the job seeker and the number of activities of the job seeker for any job offer. This makes it possible to further extract active job seekers from among job seekers who are highly suitable for the recommended job offers, and present them to the user, who is the employer.

"Any job offer" refers to any one or more job offers registered in the database, including recommended jobs and non-recommended jobs. Furthermore, "activity in relation to a job offer" includes not only actions in relation to a specific job offer, such as viewing a job offer, applying for a job offer, adding a job offer to a bookmark list, replying to a scouting document sent based on a job offer, but also actions related to all job offers, such as logging in to the recruiting support system 1 and editing one's own registered information.

Specifically, the candidate extraction unit 113 may extract candidates based on a job seeker score calculated for each job seeker. The job seeker score increases according to the magnitude of the job seeker matching index and its weighting, and also according to the magnitude of the number of activities for any job offer during a predetermined evaluation period and its weighting. The weighting of the job seeker matching index is greater than the weighting of the number of activities. This makes it possible to extract candidates that place importance on the degree of suitability while also taking into account the activity rate of the job seeker.

For example, the candidate extraction unit 113 calculates a matching score by multiplying the job seeker matching index normalized to a value between 0 and 1 by a third weighting coefficient, and an activity score by multiplying the number of activities in the evaluation period normalized to a value between 0 and 1 by a fourth weighting coefficient, and sets the sum of the matching score and the activity score as the job seeker score. Here, the third weighting coefficient is greater than the fourth weighting coefficient. The evaluation period can be, for example, within one month from the current time, within three months, etc. The normalized number of activities can be obtained, for example, by setting the number of activities of the job seeker with the largest number of activities in the evaluation period among the job seekers for whom the job seeker matching index has been calculated as a reference value, and dividing the number of activities of each job seeker in the evaluation period by the reference value. The candidate extraction unit 113 may also set the value obtained by multiplying the matching score and the activity score as the job seeker score.

Furthermore, the candidate extraction unit 113 may count the number of activities of a job seeker using a weighting that changes depending on the time. For example, the candidate extraction unit 113 may count the number of activities of a job seeker so that the further away the date and time of the activity from the present time is, the lower the count of the activity (for example, an activity that occurred more than one month ago may be counted as 0.5 times instead of 1 time).

The candidate extraction unit 113 extracts as candidates, for example, job seekers whose job seeker scores calculated from the matching scores and activity scores are equal to or greater than a predetermined threshold.

The candidate extraction unit 113 may calculate, for each job offer, a first job seeker score and a second job seeker score in which at least one of the matching index and the number of activities during the evaluation period is weighted differently, and extract, as candidates, at least one first candidate extracted based on the first job seeker score and at least one second candidate extracted based on the second job seeker score. This makes it possible to extract, for example, a first candidate who places emphasis on aptitude and a second candidate who places emphasis on activity more than the first candidate. Therefore, candidates can be presented to the user, the employer, from multiple perspectives.

The candidate extraction unit 113 extracts candidates (calculates job seeker scores) for each predetermined period. The interval at which the candidate extraction unit 113 extracts candidates is, for example, one day. The timing or interval at which the candidate extraction unit 113 extracts candidates may be the same as or different from the timing or interval at which the job offer extraction unit 112 extracts recommended jobs. The second reference information (matching index calculation model) may be updated in accordance with the extraction of candidates by the candidate extraction unit 113. In other words, the second reference information may be updated using the latest data (newly acquired information on job offers or job seekers) for each period at which candidate extraction is performed.

<Presentation Unit 114>
The presentation unit 114 is configured to present at least one recommended job offer extracted by the job offer extraction unit 112 and at least one candidate extracted by the candidate extraction unit 113 in association with each other to a recruiter.

The presentation unit 114 presents the recommended job by, for example, displaying part of the information contained in the job posting of the recommended job (job name (position name), job number (ID), conditions, etc.) on the employer terminal 20. The presentation unit 114 also presents the candidate by, for example, displaying part of the information contained in the candidate's registration information (name, organization (company name), age, current job type, job title, etc.) on the employer terminal 20.

"Presenting in correspondence" includes, for example, displaying information about recommended jobs and information about corresponding candidates side-by-side or top-to-bottom, preparing an object (e.g., a frame) for each recommended job and displaying the corresponding candidate within this object, and displaying information about the corresponding recommended job (e.g., letters or labels indicating the job name, job number, etc.) attached to the candidate.

The presentation unit 114 may present all of the multiple recommended job offers extracted by the job offer extraction unit 112, or may present a portion of the multiple recommended job offers. Furthermore, if the job offer extraction unit 112 extracts a first recommended job offer and a second recommended job offer, the presentation unit 114 may present at least one first recommended job offer and at least one second recommended job offer to the employer. Furthermore, the presentation unit 114 may highlight, by coloring, changing the font, etc., those recommended job offers that have a high expected value for closing (for example, those with an expected value for closing equal to or greater than a predetermined value, those ranked high in expected value for closing).

The presentation unit 114 may present the recommended job offers by attaching a label according to the expected value of the successful contract. This makes it possible to show the user, the job offeror, the high demand for each recommended job offer. The "label" may include, for example, a numerical value indicating the magnitude of the expected value of the successful contract or the job offer score, a graphic representation of the numerical value in the form of a graph, bar, or the like, a character or graphic indicating a rank (e.g., "Rank A", "Rank B", etc.) assigned according to the expected value of the successful contract or the job offer score, a character or graphic indicating that the expected value of the successful contract or the job offer score is equal to or greater than a predetermined value (e.g., "recommended job offer", etc.), etc.

The presentation unit 114 may present information indicating the candidate's activity status as the candidate's information. This allows the user, the employer, to be informed of the candidate's activity. "Activity status" may include, for example, the most recent login date and time, the number of views of the job posting during a specified period, the number of replies to the scouting document, the number of applications for the job posting, etc.

Figure 6 is a diagram showing an example of a recommended job posting screen RD displayed on the employer terminal 20. On the recommended job posting screen RD, multiple recommended job postings RI and multiple candidate information CI are arranged as card-shaped objects. In the example of Figure 6, three recommended job postings RI are displayed, and two pieces of candidate information CI are associated with each recommended job posting RI. Specifically, the candidate information CI is arranged to the right of the corresponding recommended job posting RI. In addition, some of the candidate information CI is affixed with a label LB indicating that the expected value of the contract is equal to or greater than a predetermined value.

By selecting any candidate information CI displayed on the recommended job posting screen RD, the user (employer) can view the registered information of the corresponding candidate (job seeker) and create a scouting document for the candidate. In other words, by operating and inputting the object of the candidate information CI, a screen for viewing the registered information of the candidate, a screen for creating a scouting document, etc. are displayed on the employer terminal 20.

The presentation unit 114 may accept input of an instruction to present recommended job offers from the user (employer) and present the recommended job offers and candidates on the employer terminal 20, or may present the recommended job offers and candidates on the employer terminal 20 in response to the occurrence of a specified event or at a specified timing. Examples of specified events include the user (employer) logging in to the recruitment support system 1, the display of a job offer management screen on the employer terminal 20, etc.

The presentation unit 114 may select and present a portion of the recommended job offers from the extracted multiple recommended job offers for each specified period. This allows new recommended job offers to be presented to the user (employer) for each specified period, thereby increasing motivation to send scout documents. The period for switching recommended job offers is, for example, one day. In other words, the presentation unit 114 may present recommended job offers on a daily basis. This can encourage the user (employer) to log in to the recruitment support system 1 every day.

Specifically, the presentation unit 114 selects the number of recommended job offers (e.g., 3) as job offers to be presented from among the multiple recommended job offers extracted by the job offer extraction unit 112, and displays the job offers to be presented on the employer terminal 20. The selection of job offers to be presented is performed, for example, based on the expected closing value or job offer score of each job offer. For example, the presentation unit 114 may select job offers to be presented from among the multiple recommended job offers in descending order of expected closing value or job offer score, and after a specified period of time, repeat the process of selecting job offers to be presented from the remaining recommended job offers in descending order of expected closing value or job offer score so that the same job offer to be presented is not selected. For example, the presentation unit 114 may present recommended job offers with the first, second, and third highest expected closing values or job offer scores on the first day, and present recommended job offers with the fourth, fifth, and sixth highest expected closing values or job offer scores on the second day.

The presentation unit 114 may also assign a probability (selection probability) to each of the multiple recommended job offers that it will be selected as a job offer to be presented, randomly select a job offer to be presented based on the selection probability, and repeat the process of selecting a job offer to be presented from the remaining recommended job offers after a specified period of time based on the selection probability again. The selection probability is set according to the expected value of contract or the level of the job offer score, and is set, for example, to 40% for the recommended job offer with the highest expected value of contract or job offer score, and 30% for the recommended job offer with the second highest expected value of contract or job offer score. The selection probability of each recommended job offer may also be set randomly.

The presentation unit 114 may select at least one job offer to be presented from each group of multiple recommended job offers extracted based on job offers scores that are weighted differently at the time of calculation. For example, the presentation unit 114 may select at least one job offer to be presented from each of a first group consisting of multiple first recommended job offers (job offers extracted with the first job offer score), a second group consisting of multiple second recommended job offers (job offers extracted with the second job offer score), and a third group consisting of multiple third recommended job offers (job offers extracted with the third job offer score). In other words, the presentation unit 114 may select and present the first recommended job offer, the second recommended job offer, and the third recommended job offer from each of the first group, the second group, and the third group for each specified period.

If the number of extracted recommended job offers is equal to or less than the number of presented job offers, all recommended job offers are presented, and the presented recommended job offers are not updated until new recommended job offers are extracted by the job offer extraction unit 112. Also, if the job offer extraction unit 112 determines that there are no recommended job offers (i.e., if no recommended job offers could be extracted), the presentation unit 114 does not present the recommended job offers to the employer. Also, in accordance with the change in the displayed recommended job offers, the candidates corresponding to the recommended job offers are also changed.

The presentation unit 114 may select and present a portion of candidates from the extracted candidates for each specified period. This allows new candidates to be presented to the user (employer) for each specified period, thereby increasing motivation to send scouting documents. The period for switching candidates is, for example, one day. In other words, the presentation unit 114 may present candidates on a daily basis.

The period for switching candidates may be the same as the period for switching recommended job offers, or may be different. For example, the presentation unit 114 may change only the candidates presented in association with the recommended job offers without changing the recommended job offers presented by making the period for switching candidates shorter than the period for switching recommended job offers.

Specifically, the presentation unit 114 selects the number of candidates to be presented (e.g., 2) from among the multiple candidates extracted by the candidate extraction unit 113 as candidates to be presented, and displays the candidates to be presented on the employer terminal 20. The selection of candidates to be presented is performed, for example, based on the job seeker matching index or job seeker score of each candidate. For example, the presentation unit 114 may select candidates to be presented from among the multiple candidates in descending order of the job seeker matching index or job seeker score, as in the case of recommended job offers, and after a specified period of time, repeat the process of selecting candidates to be presented from among the remaining candidates in descending order of the job seeker matching index or job seeker score so that the same job offer to be presented is not selected. Also, as in the case of recommended job offers, the presentation unit 114 may assign a probability of being selected as a candidate to be presented (selection probability) to each of the multiple candidates, randomly select candidates to be presented based on the selection probability, and repeat the process of selecting candidates to be presented from among the remaining candidates after a specified period of time again based on the selection probability.

The presentation unit 114 may exclude from the candidates to be presented those candidates extracted by the candidate extraction unit 113 who have received a scouting document from the user (employer) based on the corresponding recommended job offer within a specified period (e.g., within one month from the present time). This prevents candidates who have already received a scouting document from being presented to the employer.

The presentation unit 114 may switch only the candidates to be presented at predetermined intervals, without switching the recommended job offers to be presented at predetermined intervals (i.e., keeping the recommended job offers to be presented fixed until new recommended job offers are extracted).

The job extraction unit 112 may extract recommended job offers, and the candidate extraction unit 113 may extract candidates in accordance with the period for switching the recommended job offers and candidates presented by the presentation unit 114. In other words, the timing for switching the recommended job offers and candidates presented may be synchronized with the timing for extracting the recommended job offers and candidates. For example, if the presentation unit 114 switches the recommended job offers and candidates on a daily basis, the job extraction unit 112 and the candidate extraction unit 113 extract the recommended job offers and candidates (calculate the job offer score and job seeker score) once a day. The presentation unit 114 selects the recommended job offers for presentation and the candidates for presentation from the recommended job offers and candidates extracted on a daily basis.

In this way, when switching the recommended job offers and candidates to be presented while sequentially extracting the recommended job offers and candidates, the presentation unit 114 may select the recommended job offers to be presented and the candidates to be presented from among the extracted recommended job offers and candidates that have not been presented within a specified period (e.g., the previous day or the most recent week) in the above-described procedure. In other words, the presentation unit 114 may perform a process to remove the recommended job offers and candidates presented within a specified period from the extracted recommended job offers and candidates. In this case, the job offer extraction unit 112 and the candidate extraction unit 113 may additionally extract new recommended job offers and candidates according to the number of recommended job offers and candidates removed.

The presentation unit 114 may display a label according to the expected value of a contract for job postings registered by the user (employer) in situations other than when the recommended job postings are presented. For example, the presentation unit 114 may assign a label to a list of job postings registered by the user (employer). This allows the user (employer) to know which job postings are in high demand (high expected value of a contract) from among the job postings they have.

Figure 7 is a diagram showing an example of a job listing screen LD displayed on the employer terminal 20. The job listing screen LD displays a list of information (status) for multiple job listings. The job listing screen LD displays all job listings registered by the employer, or job listings extracted using search criteria (i.e., as search results). In the example of Figure 7, a label LB indicating a high expected value of success is added to the right of the position name of a job listing with a closing expectation value equal to or greater than a predetermined value.

<Scout Document Management Department 115>
The scout document management unit 115 is configured to create a scout document based on a job offer and to transmit the scout document to a job seeker. The scout document management unit 115 receives a selection input from the recruiter terminal 20 of the job offer from which the scout document is to be created and the job seeker to whom the scout document is to be transmitted, and creates a scout document. For example, when an arbitrary candidate is selected on the recommended job offer presentation screen RD of FIG. 6, the scout document management unit 115 receives the candidate as the job seeker to whom the scout document is to be transmitted, and the recommended job offer corresponding to the candidate as the source of the scout document. The scout document may be sent to the job seeker together with the job offer advertisement by attaching the job offer advertisement of the recommended job offer. Alternatively, the contents of the recommended job offer may be inserted into the scout document.

The scouting document may be created by the user (job seeker) inputting characters, or by using a learning model that has been trained to take a job posting as input and output a scouting document, or by a combination of these.

The created scout document is sent to the job seeker as a scout email. The scout document management unit 115 manages the number of scout emails that can be sent for each recruiter. The recruiter sends the scout document to the job seeker by consuming their own stock of scout emails. If the stock of scout emails required to send the scout document is insufficient, the scout document management unit 115 accepts the purchase of scout emails (a charge to increase the stock of scout emails) from the recruiter.

The scout document management unit 115 may provide the user (employer) with a stock of limited scout emails that can be sent only to candidates presented by the presentation unit 114, free of charge, at a fixed quantity (e.g., 1 to 10 emails) for each specified period. The specified period for which the limited scout emails are provided is, for example, the same as the period for which the recommended jobs and/or candidates presented by the presentation unit 114 are updated. For example, by setting the specified period to one day, the user (employer) can be encouraged to log in to the recruitment support system 1 and send scout documents every day. In addition, the expiration date for the limited scout email is set to the time when the next limited scout email is provided, and for example, if the limited scout email is provided every day, the expiration date for use is the same as the day it is provided.

In addition, when a user (employer) sends a scout email to a candidate presented by the presentation unit 114, the scout document management unit 115 may send the scout email to the candidate without consuming the recruiter's stock of scout emails.

<Artificial Intelligence Department 120>
The artificial intelligence unit 120 is configured to receive input from each functional unit and return the instructed output. The artificial intelligence used by the server device 10 in each functional unit may be a common one, or may be prepared individually for each functional unit.

The artificial intelligence unit 120 may be an AI (Artificial Intelligence) equipped with a transformer including a GPT (Generative Pretrained Transformer, including GPT-1, GPT-2, and GPT-3), a BERT (Bidirectional Encoder Representations from Transformers), a BART (Bidirectional and Auto-regressive Transformer), and a language model such as a recurrent neural network (RNN), and may include a generative AI.

The language model is an example of a learning model based on a machine learning algorithm. Specific examples of machine learning algorithms include nearest neighbor methods, naive Bayes methods, decision trees, support vector machines, and deep learning using neural networks. The artificial intelligence unit 120 can apply the above algorithms as appropriate.

The artificial intelligence unit 120 may have a trained model constructed by a learning method such as supervised learning, unsupervised learning, or self-supervised learning. In supervised learning, machine learning is performed using training data (training data). The training data consists of a pair of input data for learning and output data (correct answer data). In addition, the language model may not only be one trained for a specific task, but also a general-purpose model that can be used for a wide range of tasks.

The artificial intelligence unit 120 may be a general-purpose natural language processing learning model such as a large-scale language model (LLM) that has learned a huge amount of data as the artificial intelligence. Such a general-purpose learning model includes a language model that can handle various tasks without fine tuning by one-shot learning, few-shot learning, etc. Furthermore, the general-purpose learning model may be configured to be capable of handling various tasks by zero-shot learning as well. The artificial intelligence used in each functional unit of the control unit 11 may be a separate learning model, or may be a common general-purpose learning model.

The learning models included in the artificial intelligence unit 120 (learning models used in each functional unit, such as the expected value calculation model, matching index calculation model, etc.) can undergo additional learning as transfer learning or fine tuning. For example, each time new job seeker registration information and new job posting registration occur, the artificial intelligence unit 120 may perform additional learning and fine tuning using these as new teacher data. This improves the accuracy of the information output from the learning model.

The learning model included in the artificial intelligence unit 120 may be a learning model (distilled model) obtained by knowledge distillation using the original learning model. In knowledge distillation, a trained model such as a large-scale language model is used as a teacher model, and the parameters of the student model are adjusted so that the output loss (Soft Target Loss) of the student model (distilled model) relative to the output (Soft Target) of the teacher model is small, thereby learning the student model, and the student model becomes a distilled model. In addition, the student model may be trained so that the output loss (Hard Target Loss) of the student model relative to the correct label (Hard Target) of the teacher data (combination of input data and output data of the learning model) is small. Compared to the original learning model (teacher model), the distilled model has a smaller number of parameters and a smaller processing load while having performance close to the learning model. Therefore, by using the distilled model, the cost of the recruitment support system 1 can be reduced.

For example, the expected value calculation model or the matching index calculation model may be a distilled model trained using a combination of input data and output data in a large-scale language model as training data. In addition, when the recruitment support system 1 is introduced, a large-scale language model may be used as the expected value calculation model or the matching index calculation model, and when training data from the large-scale language model is accumulated, the distilled model obtained by knowledge distillation using the training data may be used as the expected value calculation model or the matching index calculation model.

<Display section>
The display unit 211 of the recruiter terminal 20 and the display unit 311 of the job seeker terminal 30 each display a screen indicated by the screen data transmitted from the server device 10 .

<Operation acquisition section>
The operation acquisition unit 212 of the recruiting party terminal 20 accepts operations by a user (recruiter) who uses the recruiting party terminal 20. The operation acceptance unit 312 of the job seeker terminal 30 accepts operations by a user (job seeker) who uses the job seeker terminal 30.

3. Information Processing Method This section describes an information processing method of the server device 10. This information processing method is executed by a computer, with each unit of the server device 10 acting as each step.

This information processing method includes a job offer extraction step, a candidate extraction step, and a presentation step. In the job offer extraction step, at least one recommended job offer is extracted from the job offers registered in the database by the employer, based on the content of the job offer and an expected value for closing calculated based on the first reference information. In the candidate extraction step, at least one candidate for the recommended job offer is extracted from the job seekers registered in the database, based on the content of the recommended job offer and a job seeker matching index calculated based on the second reference information. In the presentation step, the recommended job offer and the candidate are associated and presented to the employer.

Figure 8 is an activity diagram showing the flow of information processing (recommended job offers and candidate presentation processing) executed by the recruitment support system 1. Below, the information processing is explained along with each activity in this activity diagram.

The process of presenting recommended job offers and candidates is initiated by a specified operation by the recruiter. The specified operation is an operation by which the server device 10 initiates the process of presenting recommended job offers, and may be an indirect operation such as the recruiter logging in to the recruitment support system 1, or a direct operation such as an instruction by the recruiter to extract recommended job offers. The recruiter inputs such a specified operation on the recruiter terminal 20 (activity A101). The server device 10 receives the specified operation on the recruiter terminal 20 and extracts recommended job offers (activity A102).

After extracting the recommended job offers, the server device 10 extracts candidates corresponding to the recommended job offers (activity A103). Next, the server device 10 selects the recommended job offers and candidates to present to the employer from the extracted recommended job offers and candidates (activity A104). Furthermore, the server device 10 outputs the selected recommended job offers and candidates to the employer terminal 20 (activity A105). As a result, the recommended job offers and candidates are displayed (presented) on the employer terminal 20 (activity A106). If necessary, the employer instructs the employer terminal 20 to view the registered information of the candidates, create scouting documents, etc.

4. Operation The operation of this embodiment can be summarized as follows. In other words, recommended job offers with a high expected value of successful conclusion can be presented to the employer together with candidates who are highly suitable for the recommended job offers. Therefore, the employer, who is the user, can increase the possibility of successful conclusion of the job offer by taking an action based on the recommended job offer (e.g., sending a scout document). In addition, by presenting candidates for the recommended job offers, the employer can identify job seekers who are likely to be successful in the job offer.

Therefore, according to this embodiment, the priority of job offers to be put into operation is presented to the user, the employer, while the ease of deciding on a job offer is visualized and provided to the employer. In addition, since the personal nature of the job offers to be put into operation can be reduced on the employer's side, the criteria for deciding which job offers to put into operation can be standardized between individuals or departments. Furthermore, by presenting recommended job offers and candidates as a set, the employer can reduce the time and effort required to search for job seekers, and can find job seekers who are difficult to find through searches by the employer. For example, an employer with multiple job offers (especially an employer with many job offers) needs to prioritize multiple job offers and put these job offers into operation within the constraints of resources such as personnel within the company to find job seekers with high suitability. According to this embodiment, job offers to be put into operation can be efficiently identified, and job seekers with high suitability for the job offers to be put into operation can also be identified at the same time. This makes it possible to efficiently proceed with recruitment activities.

The above describes an embodiment of the present invention, but the present invention is not limited to this and can be modified as appropriate without departing from the technical concept of the invention.

5. Others In the above embodiment, the server device 10 performs various storage and control, but multiple external devices may be used instead of the server device 10. That is, various information and programs may be distributed and stored in multiple external devices using block chain technology or the like. In particular, the artificial intelligence unit 120 may be an external configuration of the server device 10. In that case, the artificial intelligence unit 120, which is an external configuration, is configured to receive input from each functional unit of the server device 10 and return the instructed output to the server device 10.

The aspect of this embodiment is not limited to the recruiting support system 1, and may be an information processing method or a program. The recruiting support method includes each step executed by the recruiting support system 1. The program causes a computer to execute each step of the recruiting support system 1.

It may be provided in the following ways:

(1) A recruitment support system comprising a processor, the processor being configured to execute the following steps: in a job offer extraction step, at least one recommended job offer is extracted from among the job offers registered in a database by an employer, based on the content of the job offer and a closing expectation value calculated based on first reference information, the first reference information including a correlation between the content of the job offer and the closing expectation value, which is a probability of closing the job offer for the job offer; in a candidate extraction step, at least one candidate for the recommended job offer is extracted from among job seekers registered in a database, based on the content of the recommended job offer and a job seeker matching index calculated based on second reference information, the second reference information including a correlation between the content of the job offer and the job seeker matching index, which indicates the degree of suitability of the job seeker for the job offer; and in a presentation step, the recommended job offer and the candidate are associated and presented to the employer.

(2) In the recruitment support system described in (1) above, the first reference information is an expectation calculation model that has been trained to be able to input the contents of a job offer and output the expected value of the deal, and in the job offer extraction step, the contents of the job offer are input to the expectation calculation model, and the expectation calculation model is caused to output the expected value of the deal. A recruitment support system.

(3) In the recruitment support system described in (2) above, the expected value calculation model is learned using the contents of multiple job offers and whether or not these job offers have been concluded within a predetermined period of time.

(4) In the recruitment support system described in any one of (1) to (3) above, in the job extraction step, job offers whose expected value of closing is equal to or greater than a predetermined threshold are extracted as the recommended job offers.

(5) In the recruitment support system described in any one of (1) to (4) above, in the job offer extraction step, the recommended job offers are extracted from the job offers registered in the database by the recruiter based on the expected value of the job offer and the history of sending scouting documents based on the job offers to job seekers.

(6) In the recruitment support system described in (5) above, in the job extraction step, the recommended job offers are extracted based on a job offer score calculated for each job offer, the job offer score increases according to the magnitude of the expected value of closing and its weighting, and decreases according to the magnitude of the number of times the scout document is sent in a predetermined period and its weighting, and the weighting of the expected value of closing is greater than the weighting of the number of times it is sent.

(7) In the recruitment support system described in (6) above, in the job extraction step, a first job score and a second job score in which at least one of the expected value for closing and the number of submissions is weighted differently for each job, and at least one first recommended job extracted based on the first job score and at least one second recommended job extracted based on the second job score are extracted as the recommended job offers, and in the presentation step, the first recommended job offer and the second recommended job offer are presented to the employer.

(8) In the recruitment support system described in any one of (1) to (7) above, the second reference information is a matching index calculation model that is trained to be able to input the contents of a job offer and output the job seeker matching index for each job seeker, and in the candidate extraction step, the contents of the recommended job offer are input to the matching index calculation model, and the matching index calculation model is caused to output the job seeker matching index.

(9) In the recruitment support system described in (8) above, the matching index calculation model is a learning model that learns the relationship between the contents of multiple job offers and job seekers who have a history of actions related to the job offers, and the recruitment support system outputs the job seeker matching index as the degree of similarity between the input job offer and a job seeker who has a history of the actions related to a similar job offer.

(10) In the recruiting support system described in (9) above, the action is receiving a scouting document or passing a screening process.

(11) A recruitment support system according to any one of (1) to (9) above, wherein in the candidate extraction step, the candidate is extracted from among job seekers registered in a database based on the job seeker matching index of the job seeker and the number of activities of the job seeker for any job offer.

(12) In the recruitment support system described in any one of (1) to (11) above, in the presentation step, a portion of the recommended job offers is selected from the extracted plurality of recommended job offers for each specified period and presented.

(13) A recruitment support system according to any one of (1) to (12) above, wherein in the presentation step, a portion of the candidates is selected from the extracted candidates for each specified period and presented.

(14) The recruiting support system according to any one of (1) to (13) above, wherein in the presentation step, a label corresponding to the expected value of the contract is added to the recommended job offer and presented.

(15) A recruitment support method comprising steps executed by a recruitment support system described in any one of (1) to (14) above.

(16) A program that causes a computer to execute each step of the recruitment support system described in any one of (1) to (14) above.
Of course, this is not the case.

Finally, various embodiments of the present disclosure have been described, but these are presented as examples and are not intended to limit the scope of the invention. The novel embodiments can be implemented in various other forms, and various omissions, substitutions, and modifications can be made without departing from the spirit of the invention. The embodiments and modifications thereof are within the scope and spirit of the invention, and are included in the scope of the invention and its equivalents as set forth in the claims.

1: Recruitment support system 2: Communication line 10: Server device 11: Control unit 12: Memory unit 13: Communication unit 14: Communication bus 20: Recruiter terminal 21: Control unit 22: Memory unit 23: Communication unit 24: Input unit 25: Output unit 26: Communication bus 30: Job seeker terminal 31: Control unit 32: Memory unit 33: Communication unit 34: Input unit 35: Output unit 36: Communication bus 111: Basic display control unit 112: Job offer extraction unit 113: Candidate extraction unit 114: Presentation unit 115: Scout document management unit 120: Artificial intelligence unit 211: Display unit 212: Operation acquisition unit 311: Display unit 312: Operation reception unit

Patent Citations (6)

Publication number	Priority date	Publication date	Assignee	Title
JP2013186813A	2012-03-09	2013-09-19	Ｒｅｄ－Ｚｏｎｅ株式会社	Job offer/job hunting information providing device
JP2015164022A	2014-02-28	2015-09-10	日本電気株式会社	Matching device, matching method, and program
US20200226532A1	2018-12-11	2020-07-16	Scout Exchange Llc	Talent platform exchange and recruiter matching system
US20200402014A1	2019-06-19	2020-12-24	Microsoft Technology Licensing, Llc	Predicting hiring priorities

JP7022401B1	2021-10-04	2022-02-18	株式会社フジヨシ	Human resources matching device and human resources matching program
JP7403027B1	2023-10-27	2023-12-21	株式会社ビズリーチ	Recruitment support system, recruitment support method and program
Family To Family Citations				

* Cited by examiner, † Cited by third party

Cited By (1)

Publication number	Priority date	Publication date	Assignee	Title
JP7627986B1	2024-10-08	2025-02-07	株式会社タイミー	Employment management system, employment management program, and employment management method
Family To Family Citations				

* Cited by examiner, † Cited by third party, ‡ Family to family citation

Similar Documents

Publication	Publication Date	Title
Kühl et al.	2020	Supporting customer-oriented marketing with artificial intelligence: automatically quantifying customer needs from social media
Lovaglio et al.	2018	Skills in demand for ICT and statistical occupations: Evidence from web-based job vacancies
CN104781837B	2020-09-22	System and method for forming predictions using event-based sentiment analysis
US20140358810A1	2014-12-04	Identifying candidates for job openings using a scoring function based on features in resumes and job descriptions
JP7403027B1	2023-12-21	Recruitment support system, recruitment support method and program
US20210103876A1	2021-04-08	Machine learning systems and methods for predictive engagement
US11487947B2	2022-11-01	Machine learning techniques for analyzing textual content
JP2015164022A	2015-09-10	Matching device, matching method, and program
JP7350206B1	2023-09-25	Recruitment data management system and recruitment data management method
JP7385077B1	2023-11-21	Search support system, search support method and program
JP7483170B1	2024-05-14	Information processing system, information processing method, and program
Law et al.	2020	Knowledge-driven decision analytics for commercial banking
US20240177255A1	2024-05-30	A data providing device, a data providing system, a data providing program, a data providing method, a data analysis device, a data management system, a data management method, and a data recording medium
JP7406031B1	2023-12-26	Writing support system, writing support method and program
JP5905651B1	2016-04-20	Performance evaluation apparatus, performance evaluation apparatus control method, and performance evaluation apparatus control program
WO2016189605A1	2016-12-01	Data analysis system, control method, control program, and recording medium
Marrara et al.	2017	A language modelling approach for discovering novel labour market occupations from the web
JP7488974B1	2024-05-22	Information processing system, information processing method, and program
JP7475529B1	2024-04-26	Information management system, information management method and program
JP7546181B1	2024-09-05	Recruitment support system, recruitment support method and program
US20230342693A1	2023-10-26	Methods and apparatus for natural language processing and governance
JP7535679B1	2024-08-16	Information processing system, information processing method, and program
Bouhoun et al.	2023	Information Retrieval Using Domain Adapted Language Models: Application to Resume Documents for HR Recruitment Assistance
Pendyala et al.	2022	Artificial Intelligence Enabled, Social Media Leveraging Job Matching System for Employers and Applicants
JP7505136B1	2024-06-24	Job search support system, job search support method and program

Priority And Related Applications

Priority Applications (1)

Application	Priority date	Filing date	Title

JP2024030922A	2024-03-01	2024-03-01	Recruitment support system, recruitment support method and program

Applications Claiming Priority (1)

Application	Filing date	Title
JP2024030922A	2024-03-01	Recruitment support system, recruitment support method and program

Legal Events

Date	Code	Title	Description
2024-03-01	A621	Written request for application examination	**Free format text**: JAPANESE INTERMEDIATE CODE: A621 **Effective date**: 20240301
2024-03-01	A871	Explanation of circumstances concerning accelerated examination	**Free format text**: JAPANESE INTERMEDIATE CODE: A871 **Effective date**: 20240301
2024-05-28	A131	Notification of reasons for refusal	**Free format text**: JAPANESE INTERMEDIATE CODE: A131 **Effective date**: 20240528
2024-06-13	A521	Request for written amendment filed	**Free format text**: JAPANESE INTERMEDIATE CODE: A523 **Effective date**: 20240613
2024-08-09	TRDD	Decision of grant or rejection written	
2024-08-20	A01	Written decision to grant a patent or to grant a registration (utility model)	**Free format text**: JAPANESE INTERMEDIATE CODE: A01 **Effective date**: 20240820
2024-08-28	A61	First payment of annual fees (during grant procedure)	**Free format text**: JAPANESE INTERMEDIATE CODE: A61 **Effective date**: 20240826
2024-09-04	R150	Certificate of patent or registration of utility model	**Ref document number**: 7546181 **Country of ref document**: JP **Free format text**: JAPANESE INTERMEDIATE CODE: R150

Concepts

machine-extracted

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Name	Image	Sections	Count	Query match
recruitment		title,claims,abstract,description	74	0.000
method		title,claims,abstract,description	42	0.000
extraction		claims,abstract,description	165	0.000
extract		claims,abstract,description	37	0.000
calculation method		claims,description	71	0.000
action		claims,description	53	0.000
effects		claims,description	41	0.000
biological transmission		claims,description	37	0.000
decrease		claims,description	4	0.000

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